Airline Passenger Satisfaction
Micro-Project 2

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Problem Statement

- Only 43.4% of passengers in airline.csv report being satisfied.
- Which service ratings, travel classes and arrival-delay groups show the largest differences in satisfaction?
- The analysis can help identify service areas for further investigation.

Hypothesis Formulation

- H1: Passengers who give higher service ratings are more likely to report satisfaction.
- H2: Satisfaction is lower among passengers with longer arrival delays.
- Compare group percentages and Spearman correlations. These are exploratory hypotheses, assessed descriptively without causal claims.

Acquire

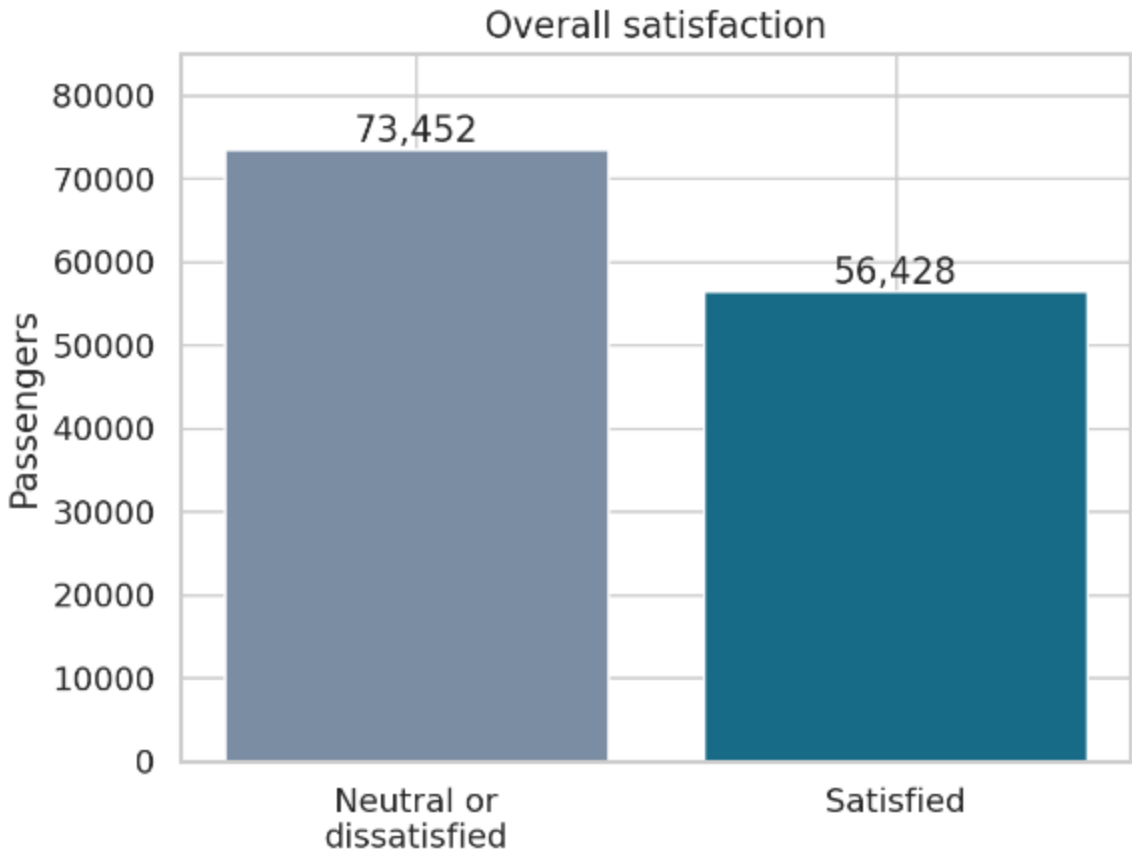
- Source: supplied airline.csv file, loaded with pandas.read_csv.
- 129,880 records and 25 original columns. Each row describes a passenger record.
- Outcome: satisfied versus neutral or dissatisfied. Fields cover passenger characteristics, travel class, service ratings and delays.
- Original publisher, survey dates and sampling method are undocumented in the supplied file.

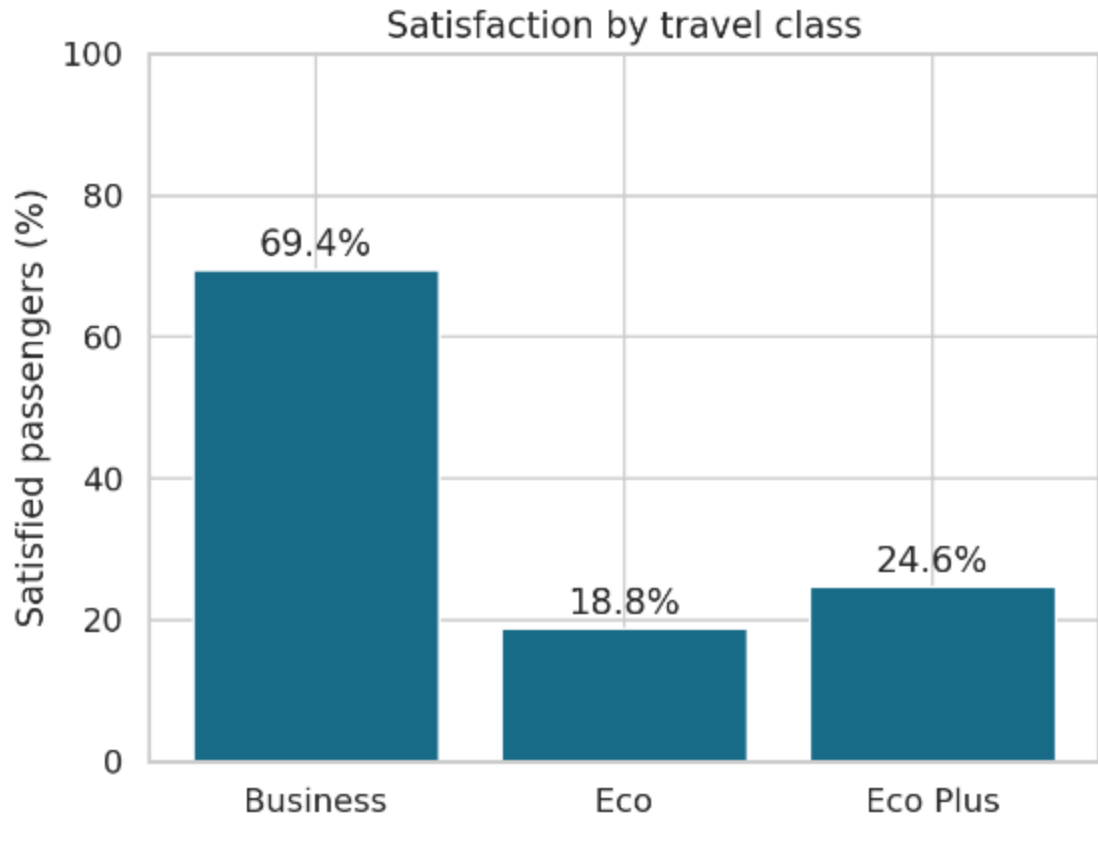
Prepare

- Remove the exported index column. Check exact duplicates and passenger IDs: 0 duplicate rows and all IDs unique.
- Retain all 129,880 records. Exclude 393 missing arrival delays only from delay analyses (n = 129,487).
- Validate age, distance, delay and rating ranges. Encode satisfied as 1 and the combined alternative as 0.
- Exclude 20,134 zero-valued service responses from rating summaries only because their meaning is undocumented.

Analyze data

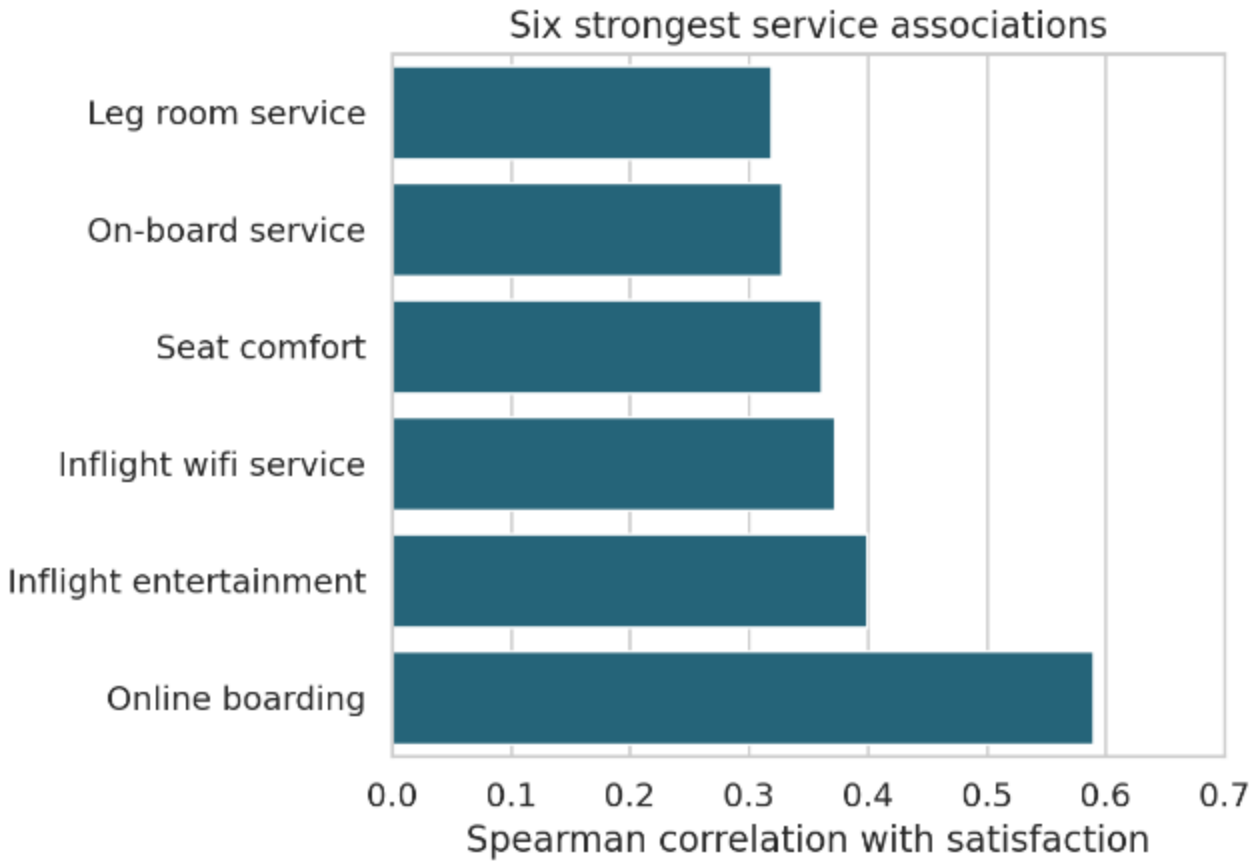
- Descriptive percentages by class, with all 129,880 records.

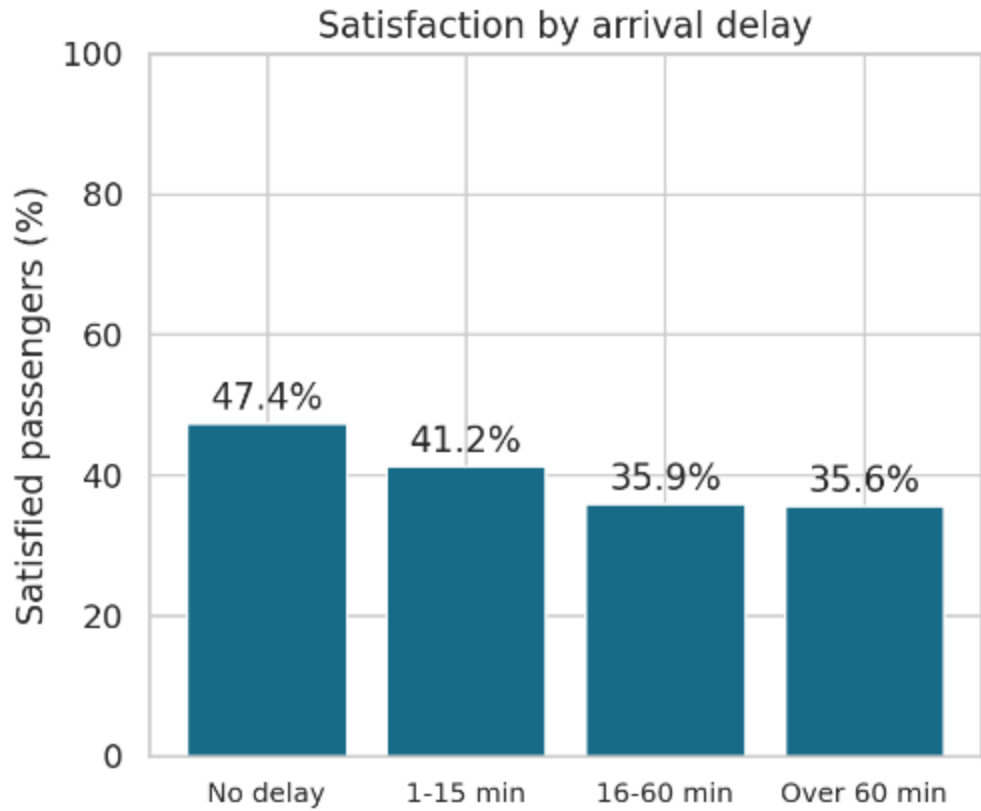




Analyze data

- Spearman correlations use nonzero service ratings. Delay comparisons use 129,487 records.





Report

- 43.4% satisfied overall. Business class: 69.4%, Eco: 18.8%, Eco Plus: 24.6%.
- Online boarding has the strongest service association with satisfaction (Spearman rho = 0.590). Mean ratings: 4.15 among satisfied passengers versus 2.71 in the combined alternative group.
- Satisfaction falls from 47.4% with no arrival delay to 35.6% with delays over 60 minutes. Both hypotheses receive descriptive support.
- Investigate boarding experience and delays further. Customer mix may confound comparisons. Unknown sampling and survey timing limit generalization.